ZEROPAY DIGITAL SERVICES LTD

United Kingdom Subsidiary of BETONIQ WEST LTD (RC 1496603, Nigeria)

LearnOffline — Technical White Paper

Offline-capable, curriculum-adaptive AI tutor for low-connectivity schools.

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Executive Summary

LearnOffline is a Zeropay Digital Services Ltd venture (UK subsidiary of BETONIQ WEST LTD) — Offline-capable, curriculum-adaptive AI tutor for low-connectivity schools.

Key statistic: 617 million children and adolescents worldwide are not achieving minimum proficiency in reading or mathematics, many without ever having access to a qualified teacher (UNESCO).

Problem Statement

Rural and low-income schools frequently face teacher shortages and unreliable internet access, both of which compound learning loss. Existing ed-tech solutions largely assume constant connectivity, excluding exactly the students who need the most support.

Technical Architecture

- Lesson packs download once over any available connection, then operate with zero ongoing internet dependency.
- Adaptive-difficulty engine adjusts question pacing per student based on demonstrated mastery, functioning entirely on-device.
- Background sync protocol uploads progress automatically the instant connectivity is detected, with conflict-safe merge logic.
- Curriculum mapping layer allows content to be aligned to a selected national or regional syllabus.

Core Data Model

Entity	Purpose
Student	Learner profile, grade level, assigned curriculum.
Lesson	Curriculum-aligned content unit, downloaded for fully offline use.
ProgressRecord	Mastery score per lesson/topic, timestamped, queued for sync.
School	Institution profile, licensing tier, enrolled students.

Security, Privacy & Compliance

- Student data stored locally on-device until sync, minimizing exposure during offline periods.
- School-level access controls ensure teachers only see their own enrolled students' progress.
- No third-party advertising or data-resale — education data used solely for learning-outcome purposes.

Feasibility & Build Status

Offline content packaging and adaptive-difficulty logic specified. Build proceeding independently on GitHub; on-device inference model live on Hugging Face Spaces with lightweight distillation for low-end hardware.

Market Opportunity

Global ed-tech market projected to exceed $404B by 2030. Target beachhead: 5,000 under-resourced rural primary schools across Nigeria.

Grant & Funder Fit

Strong fit for education innovation funding: UNESCO/UNICEF EdTech innovation funds, EU Erasmus+ digital education calls, GPE (Global Partnership for Education), and national ministry of education digital learning grants.

Funding Ask

$140,000 grant to deploy offline tutor packs across 50 rural schools, 12-month pilot with UNESCO-aligned outcome measurement.